Question 1

Objective: determine the quantity of each type of backpack to maximize profit while conforming to constraints

Part A. Decision Variables: The decision variables are quantities of collegiate backpacks produced (*c*) and mini backpacks produced (*m)*

Where

$c, m \geq 0$

Part B. Objective function:

Max Z = 32*(c)* + 24(*m)*

Part C. Constraints:

5000 square feet of material per week: $3c + 2m \leq 5000$

Sales forecasts:

$c \leq 1000$

$m \leq 1200$

Labor capacity:

$35 * 40 * 60 = 84{,}000$ labor minutes per week

$45c + 40m \leq 84{,}000$

Part. D. Full Mathematical Formulation for LP Problem

$Maximize\ Z = 32c + 24m$

Subject to:

$3c + 2m \leq 5000$

$c \leq 1000$

$m \leq 1200$

$45c + 40m \leq 84{,}000$

$c > 0$

$m > 0$

Question 2

Part A. Decision Variables

x_{11} = Number of small products produced by plant 1

x_{12} = medium by plant 1

x_{13} = large by plant 1

x_{21} = small by plant 2

x_{22} = medium by plant 2

x_{23} = large by plant 2

x_{31} = small by plant 3

x_{32} = medium by plant 3

x_{33} = large by plant 3

p = percentage of excess capacity utilized by each plant

$0 \leq p \leq 1$

Constraints:

$x11 + x12 + x13 \leq 750$

$x21 + x22 + x23 \leq 900$

$x31 + x32 + x33 \leq 450$

$12(x11) + 15(x12) + 20(x13) \leq 13000$

$12(x21) + 15(x22) + 20(x23) \leq 12000$

$12(x31) + 15(x32) + 20(x33) \leq 5000$

$x11 + x21 + x31 \leq 750$

$x12 + x22 + x32 \leq 1200$

$x13 + x23 + x33 \leq 900$

Part B. Linear Programming Model Formula

$Maximize\ Z = 300(x11 + x21 + x31) + 360(x12 + x22 + x32) + 420(x13 + x23 + x33)$

Subject to:

$x11 + x12 + x13 = 750p$

$x21 + x22 + x23 = 900p$

$x31 + x32 + x33 = 450p$

$12(x11) + 15(x12) + 20(x13) \leq 13000$

$12(x21) + 15(x22) + 20(x23) \leq 12000$

$12(x31) + 15(x32) + 20(x33) \leq 5000$

$x11 + x21 + x31 \leq 750$

$x12 + x22 + x32 \leq 1200$

$x13 + x23 + x33 \leq 900$

$x_{ij} \geq 0$

$p \geq 0$